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The Third Visual Object Tracking Segmentation VOTS2025 Challenge Results

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Abstract

The VOTS2025 is the third edition of the Visual Object Tracking Segmentation benchmark. Organised the VOT initiative, VOTS builds on 10 years of experience in organising VOT challenges. Building on the tracking setup introduced in VOTS2023, the challenge continues to integrate short-term and long-term tracking, as well as single-target and multi-target scenarios, using segmentation masks as the sole form of target annotation. This year's benchmark features three sub-challenges. VOTS2025 and VOTSt2025, evaluate tracking of conventional objects and objects undergoing topological changes, respectively. A new addition, VOTS-RT2025, aims to foster the development of efficient tracking models by introducing constraints that highlight realtime performance. All sub-challenges adopt a consistent evaluation protocol, with VOTS-RT2025 introducing specific modifications to reflect latency-aware performance. We report and analyze results from 32 submissions. Full tracker descriptions, source code, datasets, and the evaluation toolkit are available on the project website[1].

1. Introduction

The Visual Object Tracking (VOT) initiative was established over a decade ago to address the lack of standardization in performance evaluation for visual object tracking. Since the inaugural VOT2013 challenge [29], the initiative has grown into the principal benchmarking effort in the field, driving progress through the systematic design of challenges, datasets, evaluation protocols, and toolkits, in collaboration with the research community through annual workshops at major vision conferences. Over the years, VOT has launched a series of challenges considering various tracking aspects, such as short-term [29], thermal [30], real-time [25], long-term [24], RGB-D and RGB-T [22], segmentation tracking [21] and tracking of objects that break apart [28] — each coupled with carefully designed datasets and performance evaluation protocols.

A wealth of related activities emerged in parallel to VOT. DAVIS [43], YouTube-VIS [57], OVIS [44], BDD100K MOTS [62], KITTI MOTS [52], and OVTrack [33], have provided valuable datasets for video instance segmentation and open-vocabulary tracking. Challenges like LVIS [14], VIPSeg [37], and MOSE [9] contributed to multi-object and long-tail tracking evaluation. Different from these, the VOT initiative has focused on the traditional tracking community interested in tracking any object or part specified by a single training example in the first frame.

By evolving the general object tracking benchmarks, VOT has often captured key methodological shifts at their earliest stages, before their mainstream adoption. Early versions of discriminative correlation filters like KCF [15] gained visibility in VOT2014; Siamese trackers such as SiamFC [3] and SiamRPN [32] stood out in VOT2016 and VOT2018 with their realtime tracking potential; deep discriminative correlation filters like DiMP [4] dominated VOT2019; and Transformer-based trackers such as TransT [6] were top-ranked in VOT2021. The memory-driven segmentation tracker AOT [59] demonstrated remarkable robustness in VOT2022, where segmentation-based trackers outperformed bounding-box-based methods, prompting a paradigm shift in the field. Most recently, the VOTS2024 results presentation [27] reported the possible rise of video object segmentation foundation models, such as SAM2 [45] (preliminary released in summer 2024), signaling the next wave of tracking architectures.

The evolution of the initially bounding-box-based VOT benchmarks culminated in VOTS2023 [26], which unified general object tracking by segmentation in a single tracking setup. It consolidated previously separate short-term, long-term, single-target, and multi-target tracking, under a single segmentation-based paradigm. Alongside this unification, a new set of performance measures and an evaluation protocol was developed to analyze the different tracking performance aspects.

In the thirteenth edition of the VOT challenges (and the third VOTS challenge edition), VOTS2025 expands the scope of the segmentation-based tracking setup. In addition to the standard VOTS2025 and the topological tracking challenge VOTSt2025, a new sub-track, VOTS-RT2025, is introduced to promote efficiency and real-time capability in segmentation-based trackers. All sub-challenges follow a consistent evaluation methodology, with VOTS-RT2025 incorporating latency-sensitive extensions. This paper presents VOTS2025, organized in conjunction with the ICCV2025 Visual Object Tracking and Segmentation Workshop, and the results obtained. In the following, we overview the challenge and participation requirements.

1.1. The VOTS2025 challenge

The evaluation toolkit and the datasets were provided by the VOTS2025 organizers. The challenge opened on May 13 and closed on June 22 2025. The results, along with the winners were disclosed in early July 2025. The analysis of the results were presented at the ICCV VOTS2025 workshop on October 20th, 2025. The *VOTS2025 Benchmark* opened[2] with a continually updated leaderboards to facilitate tracker development in the post-challenge period. Three challenges were organized in VOTS2025:

- **VOTS2025 challenge** was a continuation of the VOTS2023 and VOTS2024 challenges, with the task to

[1] https://www.votchallenge.net/vots2025/

[2] https://www.votchallenge.net/vots2025/leaderboards.html

track one or more general targets over short-term or long-term sequences by segmentation.

- **VOTS-RT2025 challenge** was a real-time version of the VOTS2025 challenge, i.e., the tracker is required to report target location at real time speed.
- **VOTSt2025 challenge** was a continuation of the VOTSt2024 challenge, with the task of tracking objects undergoing topological transformations.

Participants integrated their trackers into the VOTS2025 evaluation kit, which provides the protocols and datasets, and automatically runs standardized experiments. Trackers were registered on the evaluation server by the participants, which enabled them to submit the results for server-side evaluation. Only initialization frame annotations were public; ground truth for the remaining frames was kept sequestered. Each participant was limited to 10 evaluations, with a maximum of one per day.

Submissions included both novel and previously published trackers, as well as significantly modified third-party methods. Each was accompanied by a brief abstract and a design questionnaire. These descriptions form the basis for the tracker summaries in Appendix A, Appendix B and Appendix C.

Participants with sufficiently well performing submissions, i.e. exceeding the STARK-multi [26] for VOTS2025 and AOTPlus [50] for VOTSt2025, who contributed the text for this paper and agreed to make their tracker code publicly available on the VOTS page were offered co-authorship of this results paper. The committee reserved and exercised the right to disqualify any tracker that, by their judgment, attempted to cheat the evaluation protocols.

The trackers submitted to last-year's VOTS2024 could not be re-submitted to the same VOTS2025 sub-challenge, since the same dataset and the evaluation protocol are used. However, their submission was encouraged to the other sub-challenges. The VOTS committee members could participate in the challenge, but could not compete for the winner title. All co-authors of this paper, including the VOTS2025 committee members and the tracker authors were required to specify the division of work.

Validation and test splits of popular tracking datasets were *not allowed for training* the trackers. These include OTB [54], VOT, ALOV [49], UAV123 [38], NUSPRO [31], TempleColor [34], AVisT [40], LaSOT-val [11], LaGOT [36], GOT10k-val/test [16], TrackingNet-val/test [39], and TOTB [12]. The training split of any dataset is allowed (including LaSOT-train, TrackingNet-train, YouTubeVOS, COCO, etc.). To include transparent objects, the Trans2k [35] dataset[3] was suggested.

The remainder of this report is structured as follows. Section 2 overviews the three challenges and performance evaluation protocols, the results are discussed in Section 3, and conclusions are drawn in Section 4. Short descriptions of the tested trackers and division of work are available in the Appendix A, Appendix B, and Appendix C.

[3] https://github.com/trojerz/Trans2k

2. The VOTS2025 challenges and protocols

The VOTS2025 challenge featured three sub-challenges. All challenges used the unified performance evaluation methodology proposed in VOTS2023 [26] with slight modification in the VOTS-RT2025. We first review the methodology in Section 2.1 and the individual challenges in Section 2.2, Section 2.3 and Section 2.4, respectively.

2.1. The VOTS performance evaluation protocol

The tracker is initialized on all targets in the first frame. In each subsequent frame, it must report the location of every visible target. Specifically, a segmentation mask is required for each visible target, while absent targets must be labeled as "not present." The objective is to consistently track each target initialized in the first frame. The tracker will, in general, have a very low performance, if it drifts to the background or to a different target.

A notion of tracking success was defined in VOTS2023 [26], from which the tracking quality plot can be obtained, replacing the widely-used success plot [54]. Note that, in contrast to the success plot [54], the tracking quality plot also accounts for the long-term tracking properties (see [26] for details). The VOTS primary performance measure is thus the tracking quality Q, which is related to the area under the curve in the tracking quality plot.

In addition, several secondary performance measures were proposed to aid tracking analysis. The first two are localization *accuracy* and *robustness*. The accuracy (Acc) is defined as the sequence-normalized average overlap over successfully tracked frames (i.e., with IoU > 0), while tracking robustness (Rob) is defined as the percentage of successfully tracked frames with visible target, averaged over all sequences. Following our prior work [23], the tracker performance on frames with visible target is summarized by the AR plots [21], with the top-right position indicating the better performance. Three additional secondary measures were proposed to quantify different tracking properties. These are *Not-Reported Error* (NRE), which measures the percentage of frames where the tracker incorrectly reported the target as absent; *Drift-Rate Error* DRE measures the percentage of frames where the tracker drifted off the target; while *Absence-Detection Quality* ADQ, gives the percentage of frames with target correctly predicted as absent.

2.2. The VOTS2025 challenge

The VOTS2025 subchallenge was a continuation of the VOTS2023 [26] and VOTS2024 [28]. This challenge con-

sidered tracking by segmentation of one or more targets, whose topology remained constant (i.e., classical objects or their parts). The VOTS2023 dataset was employed, which is composed of 144 sequences, and contains 341 targets in total. The average length of a sequence is ≈2000 frames (min = 63, max = 10700, median = 1810). The number of targets in a sequence ranges from 1 to 8 (median = 2, mean = 2.37). Of the 144 sequences, 93 contain a target which at least once leaves the field of view and then returns. Of the 341 targets, this happens with 168. In cases when the target leaves and returns to the field of view, the median number of absences is 3, with maximum being 23. The median absence length in terms of frame number is 18. All targets are annotated by segmentation masks. The ground truth masks are publicly available only for initialization frames, while the evaluation ground truth is sequestered on the evaluation server and is not publicly available.

Winner identification protocol. The VOTS2025 challenge winner was identified as the tracker that obtained the top tracking quality score Q among all valid submissions. The winners are required to make the tracker code publicly available before the VOTS2025 workshop.

2.3. The VOTS-RT2025 challenge

The VOTS-RT2025 subchallenge was an extension of the VOTS2025 challenge. The task was the same, with the following differences. The toolkit provides a frame and the tracker is required to report the position of all targets at 20fps speed. If the positions are not received by the toolkit in time, the toolkit takes the last reported positions as the approximations for the current frame and proceeds to the next frame. To homogenize the evaluation on hardware side, the participants were encouraged (but not required) to run their tracker on a computer with specifications defined by the VOTS-RT2025 committee (detailed below). All trackers submitted to the VOTS-RT2025 were also evaluated (by the toolkit on participant's side) on the VOTS2025 for an in-depth performance analysis.

Winner identification protocol. The VOTS-RT2025 committee re-run the submitted trackers on the same machine with the following specifications: NVIDIA RTX 4500 Ada GPU with 24GB of VRAM, Intel i9-13900K CPU and 128GB of RAM, Debian 6.1.119 (OS) CUDA 12.4.0. The specifications were made known to the participants in advance. The tracker with the highest obtained Q-score in the re-run was the challenge winner.

2.4. The VOTSt2025 challenge

The VOTSt2025 subchallenge is a continuation of the VOTSt2024 challenge. It targets complex object transformations over time and builds on the dataset introduced in [50] (CVPR2023). It comprises 713 high-resolution videos featuring tracked objects undergoing significant changes in appearance, shape, or state, e.g. an egg cracking. Additional challenges such as occlusions and long-term disappearances are also common.

The videos (as a set) averages 21.2 seconds in length, with instance masks annotated at 5 FPS, totaling 75,547 labeled frames. Participants of the VOTSt2025 challenge were provided frames at 10 FPS for better temporal consistency, though only labeled frames were used for evaluation. Ambiguous regions—due to motion blur or semi-transparency—were annotated with an "Ignore" label to ensure fairness. On average, 2.3 masks are annotated per frame.

The dataset is divided into 572 training, 70 validation, and 71 test videos. Annotations for the training and validation sets are publicly available, while only the first-frame masks are provided for the test set used in the challenge. Submissions are evaluated using the standard VOTS metrics (see Section 2.1), with a single modification: following [50], scores are computed over the final 25% of frames. This emphasizes robustness to transformations—which typically occur later in the sequence—rather than general tracking performance, which is the focus of the VOTS subchallenge.

Winner identification protocol. The VOTSt2025 challenge winner was identified as the tracker that obtained the top tracking quality score Q among all valid submissions.

3. The challenge results

This section summarizes the trackers submitted, results analysis and winner identification for each of the three VOTS2025 sub-challenges.

3.1. The VOTS2025 challenge results

A total of 21 trackers was submitted to the VOTS2025 evaluation server. After removing the duplicate, near-duplicate and incomplete submissions, 15 valid entries remained in the VOTS2025 challenge: S3-DAM4SAM (A.8), DAM4SAM (A.1), DAMTrack (A.6), SAMPT (A.14), DQASOTg (A.4), DQASOT (A.12), DAM4SAM2-SAM1 (A.5), DQASOTb (A.2), MMF_Tracker (A.13), HQ-DMAOT (A.11), DMAOT_TTA (A.9), edgesam (A.7), EVPTrack (A.10), DeSAMTrack (A.3), and MIEM-HSE_ALPHACHIP_Qwen_VL (A.15). Each submission included the link to the source code to allow verification of the results if required. The authors agreed to make the source codes publicly accessible by the VOTS2025 workshop date. After the deadline, the MMF_Tracker was withdrawn by the authors (see a note in A.13).

In addition, the VOTS2025 committee included 5 baselines: s3_Track [28] (the winner of the VOTS2024 challenge), SAM2 [45] (the VOTS2024 highlight tracker and foundation of many this-year's submissions), DMAOT [26] (the winner of the VOTS2023 challenge), MSAOT [21]

Tracker	Q	A	R	NRE	DRE	ADQ
S3-DAM4SAM	0.751①	0.800	0.911①	0.045②	0.043①	0.822①
DAM4SAM	0.711②	0.790	0.868②	0.060	0.072②	0.728
DAMTrack	0.706③	0.786	0.861	0.060	0.078	0.727
SAMPT	0.704	0.786	0.860	0.059③	0.082	0.722
DQASOTg	0.702	0.809②	0.836	0.084	0.080	0.759②
DQASOT	0.702	0.809①	0.836	0.084	0.080	0.756③
DAM4SAM2-SAM1	0.701	0.778	0.864③	0.060	0.076	0.726
DQASOTb	0.690	0.805③	0.829	0.093	0.078	0.724
MMF_Tracker†	0.689	0.776	0.853	0.068	0.079	0.735
HQ-DMAOT	0.637	0.754	0.788	0.139	0.074③	0.749
DMAOT_TTA	0.629	0.755	0.785	0.139	0.076	0.724
edgesam	0.580	0.730	0.741	0.158	0.101	0.737
EVPTrack	0.564	0.722	0.791	0.060	0.149	0.310
DeSAMTrack	0.558	0.693	0.746	0.129	0.124	0.627
MIEM-HSE_ALPHACHIP_Qwen_VL	0.366	0.585	0.564	0.037①	0.399	0.235
S3_Track°	0.722	0.784	0.889	0.070	0.041	0.781
SAM2°	0.642	0.789	0.751	0.190	0.059	0.837
DMAOT°	0.636	0.751	0.795	0.139	0.066	0.731
MSAOT°	0.550	0.698	0.767	0.096	0.137	0.470
starkmulti°	0.297	0.441	0.594	0.220	0.186	0.486

Table 1. Numerical results for VOTS2025 challenge: Tracking quality (Q), accuracy (A), robustness (R), not-reported error (NRE), drift-rate error (DRE) and absence-detection quality (ADQ). The VOTS2025 baselines are denoted by $(\cdot)^\circ$ and placed at the bottom of the table, as they are not considered in ranking of submissions, while $(\cdot)^\dagger$ denotes a tracker withdrawn by the authors after the deadline.

(winner of VOT2022 and foundation of many top trackers since VOTS2023) and StarkMulti [26] (the baseline introduced in VOTS2023). Thus, 20 trackers were evaluated in the VOTS2025 challenge.

In the following we summarize the statistics of the submitted trackers (excluding the five baselines) and refer the reader to the Appendix A for the trackers short descriptions. 80% of submissions were based on the SAM2 [45] tracker and 33% on its recent extension DAM4SAM [51]. 67% directly predicted the target segmentation mask, while the rest (4 in total) were bounding box trackers, of which all but one applied bounding box post-processing to segment the target. 27% of submissions applied classic target templates, while 67% applied memory banks (e.g., as in MSAOT [21] and SAM2 [45]). 33% of the trackers localized the target in a constrained search region, while 67% applied image-wide target localizatioin. Most trackers applied ViT [10] and/or Hiera [45] backbones.

The results are summarized in the tracking quality plots and AR plots (Figure 1), and in Table 1. The top four trackers that stand out according to the primary tracking quality score (Q) are: S3-DAM4SAM (A.8), DAM4SAM (A.1), DAMTrack (A.6), and SAMPT (A.14). The top-performer S3-DAM4SAM is the VOTS2024 winner s3_Track [28] extension and proposes a robust semantic-aware and query-enhanced video object segmentation method. A fusion block integrates the ViT's CLS token with multi-scale CNN features to extract detailed semantic features at multiple scales. A discriminative query representation module captures the local representation of the targets for distractor robustness. The target queries are updated with the most distinctive features (i.e., channels) of the target object. The tracker combines DAM4SAM [51] and Cutie [7] with encoder sharing, and proposes a method to combine the outputs of the two trackers for increased robustness and accuracy.

The second-best performing tracker is DAM4SAM [51], which is a recent extension of SAM2.1 [45]. In particular, this tracker proposes a training-free drop-in memory for SAM2.1, which explicitly addresses the distractors and robust target model updating. The third-best tracker is DAMTrack, which combines DAM4SAM [51] and DMAOT [26]. Specifically, a Kalman filter-based track selection method is proposed to select the best output of the two base trackers. Closely following DAMTrack is the fourth-best tracker SAMPT. This tracker combines the segmentation-based DAM4SAM [45] tracker with the recent point tracker CoTracker [17]. The point-tracker is leveraged for the SAM2.1 mask selection as well as for point prompting in mask refinement in cases when the output masks are scored low.

S3-DAM4SAM obtains the highest tracking quality

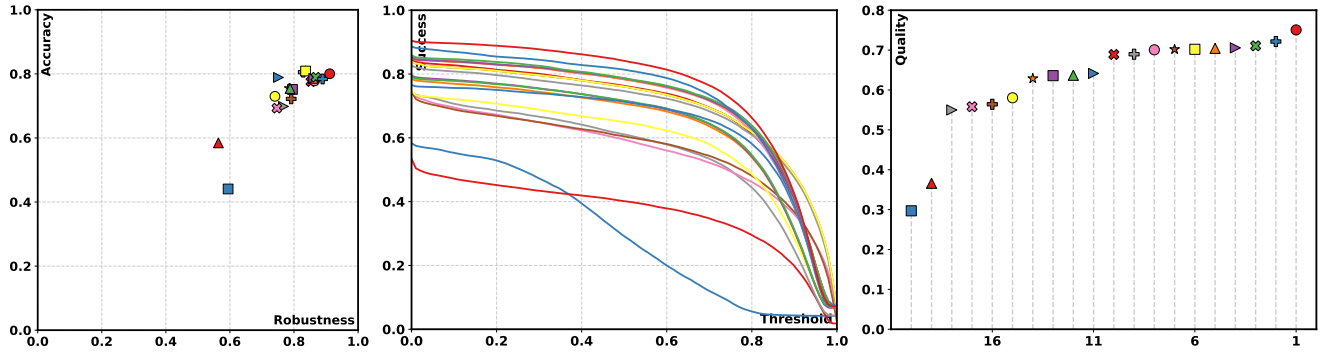


Figure 1. The VOTS2025 AR-plots (left), tracking quality Q-plots (center) and trackers ranked according to Q score (right).

(Q=0,751), which is a 5.6% improvement over the second-best entry DAM4SAM and outperforms it in both accuracy and robustness. The AR plot (Figure 1) indicates that S3-DAM4SAM strikes an excellent balance between accurate target segmentation (Acc=0.800) and very good robustness (Rob=0.911) – the latter indicates that this tracker successfully tracked, on average, 91% of test sequence length, which is a remarkable performance. The tracker drifted off the target in only 4% of cases (DRE=0.04), and falsely predicted the target as absent in 5% of cases (NRE=0.045). This also means that when the target was present, the tracker failed only in 9% of frames ($1 - R$), of this, 49% ($\frac{\text{DRE}}{\text{DRE+NRE}}$) of failures were due to drifting off the target and 51% ($\frac{\text{NRE}}{\text{DRE+NRE}}$) of failures were due to incorrectly reporting the target as absent. Overall, the target absence was correctly predicted in 82% of cases (ADQ=0.78).

From Figure 1 we observe that the Q-plot of S3-DAM4SAM stands out from the rest. It starts higher than for the rest and the margin persists considerably towards high thresholds. This indicates robust tracking for a range of thresholds, while the location of the elbow (high inflection point on Q-plot) towards very high thresholds implies a very good accuracy, further confirming the good balance between the localization robustness and the segmentation accuracy.

According to the robustness score, the top tracker is S3-DAM4SAM, while the top tracker in accuracy is DQASOT. The latter marginally outperformed S3-DAM4SAM by 1% (0.805 vs 0.800) but less frames successfully tracked (86% vs 91%). The tracking quality plots in Figure 1 reveal that S3-DAM4SAM produces substantially larger percentage of frames with increasingly large IoU (i.e., for thresholds up to ∼0.85), while DQASOT outperforms it for the highest IoU thresholds, which averages into the slightly more advantageous observed accuracy. We note that the top nine most robust trackers were based on the video segmentation model SAM2 [45], of these, top-four extended DAM4SAM [51]. These results suggest that the video segmentation memory-based models continue to dominate in top target localization capabilities.

The VOTS2025 committee provided a baseline tracker, StarkMulti [26] (Q=0.297), for validating the entry-level quality of the submissions. All 15 submissions outperform the this tracker. The second baseline was AOT [59] (Q=0.55), which was the VOT2022 winner. Almost all (14) submissions (93%) outperform this tracker and 10 submissions (66%) outperformed its followup and the VOTS2023 winner DMAOT [26] (Q=0.636). Over half of the submissions (60%) outperformed the recent SAM2 [45], which was spotlighted in VOTS2024 results presentation (see VOTS2024 workshop program for the presentation pdf) as a video-segmentation foundation model with a strong tracking capability. Finally, only a single submission (i.e., the VOTS2025 top performer) outperformed the VOTS2024 winner. In summary, the statistics indicate substantial improvement of this year's submissions over what was even few years ago considered a solid state-of-the-art.

The VOTS2025 challenge winner. The top tracker according to the tracking quality score Q is S3-DAM4SAM (A.8) and thus the VOTS2025 challenge winner.

3.2. The VOTS-RT2025 challenge results

A total of 6 trackers were submitted in the VOTS-RT2025 challenge: SV-DAM4SAM (B.6), DQASOTr (B.3), DAM4SAMLight (B.1), EdgeSAM (B.5), DQASOTrg (B.2), and LowFormer (B.7). Each submission included the link to the source code for VOTS committee analysis and the authors agreed to make it publicly accessible by the VOTS2025 workshop date. In addition, the VOTS2025 committee included the baseline tracker MSAOT (B.4) [21] – winner of VOT2022 and foundation of many top trackers since VOTS2023.

In the following we summarize the statistics of the submitted trackers (excluding the baseline) and refer the reader

to the Appendix B for the trackers short descriptions. 67% of submissions were based on the SAM2 [45] tracker and 33% on its recent extension DAM4SAM [51]. 83% of submissions directly predicted the target segmentation mask and applied memory banks (e.g., as in MSAOT [21] and SAM2 [45]) with image-wide target localization. The remaining method was a bounding box tracker, which localized the target in a constrained search region. Most trackers applied ViT [10] and/or Hiera [45] backbones.

The results of the self-evaluation (i.e., results obtained by participants on their own hardware) are summarized in Table 2. Since realtime performance is substantially affected by the used hardware, the comparative tracker analysis might not be faithfully reflected by the self-evaluation. The VOTS-RT2025 committee thus collaborated with the authors to re-run all the trackers on a same VOTS committee-owned machine for fair analysis. The re-analysis revealed that two trackers (DQASOTr and DQASOTrg) used sequence identity to switch between tracker variants, which is not practically useful. These trackers were thus flagged (†) in the results and excluded from top-performance competition.

The results of the re-run are shown in Figure 2 and Table 3. The top-performing tracker according to the re-evaluation Q-score is SV-DAM4SAM (B.6). This tracker extends DAM4SAM [51] by incorporating a shared feature management system. This system computes the image encoder features once per frame and shares them across all tracked objects, which speeds up the inference when multiple objects are tracked. The tracker uses SAM2.1 [45] model with Hiera-L [46] backbone. SV-DAM4SAM achieves tracking quality Q=0.626 in the committee verification process, which is a 11% performance drop compared to the performance on the main VOTS2025 challenge (Q=0.706) where the realtime constraint is not applied. A close inspection of this result reveals that the main cause for the performance drop is in 12% lower accuracy, while the robustness between the two runs remains almost unchanged (0.86 vs. 0.85). This result indicates that even if a tracker does not process every frame due to the realtime scenario, it is still able to track approximately equally long, compared to the case when realtime constraints are not applied. On the other hand, since on the "dropped" frames predictions from the past are considered, this significantly affects the accuracy.

The second-best realtime tracker (considering DQASOTr not being practically useful) is DAM4SAMLight (B.1), which is the DAM4SAM [51] with Hiera-T [46] (tiny) backbone to assure high computation efficiency. The tracker achieves tracking quality Q=0.607, which is 3% lower compared to the top-performing SV-DAM4SAM. Comparing tracking performance with the tracking quality obtained on the main experiment (i.e., without realtime constraints), the performance drop is almost negligible (0.7%). In contrast to SV-DAM4SAM, which records an evident performance drop between main and realtime experiments, similar tracking performance of DAM4SAMLight is attributed to a lightweight backbone network (Hiera-T).

In the following we compare realtime tracking performance obtained on the participant's hardware with the performance obtained on the committee's hardware. The largest drop in tracking quality (40%) among the submitted trackers is achieved by DQASOTrg, which indicates the slowest performance. Tracking quality of the top-performing SV-DAM4SAM drops by 4.6%, while DAM4SAMLight achieves even higher tracking performance, compared to their own run.

The VOTS-RT2025 committee provided a baseline tracker MSAOT [21] (Q=0.500) for validating quality of the submissions. This tracker is the VOT2022 winner and known by efficient multi-target formulation, which makes it a strong realtime baseline method. Four submissions (67%) outperform MSAOT on the realtime experiment according to the tracking quality obtained on the committee's hardware.

The VOTS-RT2025 challenge winner. The top tracker according to the tracking quality score Q obtained by the VOTS-RT2025 committee verification process is SV-DAM4SAM (B.6) and thus the VOTS-RT2025 challenge winner.

3.3. The VOTSt2025 challenge results

A total of 5 trackers were submitted to the evaluation server, including the baseline contributed by the VOTS committee: AOTPlus (C.1), DAM-RMemAOT (C.2), MMF_Tracker_T (C.3), RMem (C.4), and DQASOTt (C.5). Each submission included the link to the source code to allow verification of the results if required. The source codes are publicly accessible.

In the following we summarize the statistics of the submissions and refer the reader to the Appendix C for the trackers' short descriptions. Of the participating trackers, 3 (60%) capitalized on SAM2 foundational model for tracking, while 2 (40%) were more traditional, end-to-end trained approaches. This shows that, although SAM2 was developed for objects that do not change significantly over time, it can help improve performance in the challenging VOTSt scenario as well. The dominant network architecture used for tracking was transformers (all the submissions utilized transformers, sometimes in a combinations with a ResNet visual encoder). All of the trackers localized the targets in a single stage, and directly predicted the object mask. All submissions applied the same network for frame-to-frame localization and target re-detection, and localized the target in the entire image rather than in a constrained search

Tracker	VOTS-RT2025 (participant's hardware) Q ↑	A ↑	R ↑	NRE ↓	DRE ↓	ADQ ↑	performance on VOTS2025 Q ↑	A ↑	R ↑	NRE ↓	DRE ↓	ADQ ↑
★SV-DAM4SAM	0.656②	0.727②	0.872①	0.048②	0.080②	0.656	0.706①	0.784②	0.863①	0.061②	0.077①	0.727②
□DQASOTr†	0.659①	0.757①	0.834②	0.095	0.070①	0.717①	0.668③	0.775③	0.830②	0.093	0.077②	0.724
△DAM4SAMLight	0.584③	0.703③	0.793	0.064③	0.143	0.534	0.611	0.755	0.793	0.063③	0.143	0.479
▷EdgeSAM	0.561	0.687	0.763	0.139	0.098	0.694③	0.583	0.731	0.743	0.156	0.101	0.727①
✖MSAOT	0.461	0.576	0.748	0.096	0.157	0.470	0.549	0.697	0.768	0.095	0.138	0.465
✚DQASOTrg†	0.573	0.652	0.826③	0.080	0.093③	0.700②	0.690②	0.805①	0.829③	0.093	0.078③	0.724③
●LowFormer	0.356	0.551	0.649	0.013①	0.338	0.000	0.378	0.590	0.654	0.013①	0.334	0.000

Table 2. The VOTS-RT2025 results obtained on the participants' hardware. Tracking quality (Q), accuracy (A), robustness (R), not-reported error (NRE), drift-rate error (DRE) and absence-detection quality (ADQ). In addition, results without the real-time constraint (i.e., obtained on the VOTS2025 main challenge) are included. The (†) denotes an impractical tracker due to use of sequence id as part of the algorithm.

Tracker	VOTS-RT2025 (committee's hardware) Q ↑	A ↑	R ↑	NRE ↓	DRE ↓	ADQ ↑	VOTS-RT2025 (participant's hardware) Q ↑	A ↑	R ↑	NRE ↓	DRE ↓	ADQ ↑
★SV-DAM4SAM	0.626①	0.692	0.849①	0.057③	0.094③	0.760①	0.656②	0.727②	0.872①	0.048②	0.080②	0.656
□DQASOTr†	0.613②	0.701②	0.828③	0.083	0.088①	0.700③	0.659①	0.757①	0.834②	0.095	0.070①	0.717①
△DAM4SAMLight	0.607③	0.700③	0.833②	0.048②	0.119	0.603	0.584③	0.703③	0.793	0.064③	0.143	0.534
▷EdgeSAM	0.585	0.723①	0.753	0.154	0.093②	0.714②	0.561	0.687	0.763	0.139	0.098	0.694③
✖MSAOT	0.500	0.628	0.752	0.093	0.154	0.495	0.461	0.576	0.748	0.096	0.157	0.470
✚DQASOTrg†	0.415	0.485	0.738	0.084	0.178	0.633	0.573	0.652	0.826③	0.080	0.093③	0.700②
●LowFormer	0.351	0.549	0.654	0.013①	0.333	0.001	0.356	0.551	0.649	0.013①	0.338	0.000

Table 3. Numerical results for the VOTS-RT2025 challenge obtained on the VOTS-RT2025 committee hardware (all trackers run on same hadrware) and on participant's hardware (i.e., self-evaluation). Tracking quality (Q), accuracy (A), robustness (R), not-reported error (NRE), drift-rate error (DRE) and absence-detection quality (ADQ) are reported. The (†) denotes an impractical tracker due to use of sequence id as part of the algorithm.

region, which is not surprising given the objects breaking into parts are not spatially localized anymore. Finally, all the submissions used memory banks to track the changing appearance of the target. These statistics demonstrate that the community has converged to a fairly unified architecture for this task.

The results are summarized in the tracking quality plots and AR plots (Figure 3), and in Table 4. The top 3 trackers according to the primary tracking quality score (Q) are: DAM-RMemAOT (C.2), DQASOTt (C.5), and RMem (C.4). The top-performing methods continue the trend of single-stage tracking pipelines, with a strong emphasis on memory updating and adaptive fusion strategies. The winning submission, DAM-RMemAOT, significantly extends the RMemAOT framework (which won the challenge in 2024) by introducing a multi-level fusion architecture. At its core, DAM-RMemAOT employs a trio of AOT-based trackers — a main tracker and two temporally offset auxiliary trackers — to generate diverse predictions, which are fused using a dynamic, confidence-aware mechanism. This fusion process incorporates mask quality metrics such as IoU, consistency, and contextual cues, ensuring that the most reliable prediction is selected at each frame. In addition, the method integrates DAM4SAM [51], a SAM2-based module that provides an independent, high-fidelity prediction stream. When available, these SAM predictions are merged with the AOT ensemble output via a hybrid fusion strategy, guided by IoU-based thresholds. This hierarchical design enables DAM-RMemAOT to gracefully fall back to alternative predictions in challenging scenarios such as occlusions or rapid appearance changes.

The second-best tracker, DQASOTt, enhances a traditional end-to-end architecture with appearance-guided mask refinement, improving boundary accuracy and maintaining temporal coherence. Its unified framework balances segmentation precision with computational efficiency, making it especially effective in complex motion scenes. RMem, which placed third, retains its focus on selective memory construction, limiting redundant frame accumulation in favor of essential updates. This curated memory design has proven effective in managing long-term appearance variations. Taken together, the top submissions showcase the increasing importance of adaptive fusion, multimodal reasoning, and strategic memory selection for robust visual object tracking under transformations.

The AR plot (Figure 3) indicates that all the top-3 trackers achieve a strong combination of accuracy and robustness, with the top-performing DAM-RMemAOT getting ahead due to top accuracy, while showing similar robustness to RMem. The second place method DQASOTt also exhibits high accuracy, but falls behind on robustness compared to the highly-sophisticated, multi-level fusion of

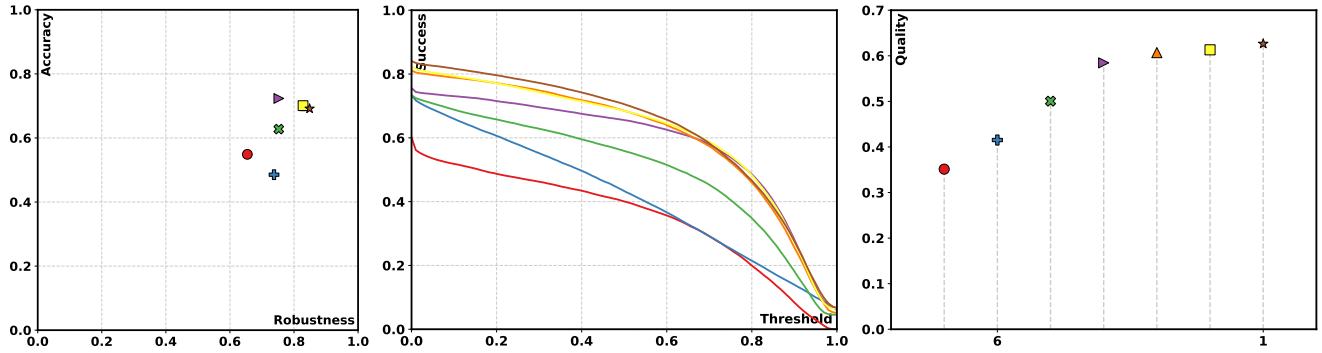


Figure 2. The VOTS-RT2025 AR-plots (left), tracking quality Q-plots (center) and trackers ranked according to Q score (right). Results obtained by re-running the trackers on VOTS2025 committee's hardware.

DAM-RMemAOT. At the same time, DQASOTt achieves top performance on auxiliary metrics such as drift (DRE) and target absence prediction (AD). This indicates that simpler tracker designs can lead to fewer spurious predictions, but more sophisticated approaches are required to achieve top performance on this challenging task.

The VOTSt2025 committee provided a strong baseline AOTPlus C.1 [50], which extends the popular AOT [59] by increasing the length of the training sequences and replacing the short-term memory module with a recurrent transformer. This tracker reached Q=0.485, which is approximately 88% of the Q-score of the top performer. Only 3 submissions outperformed the baseline, indicating that, while a non-trivial progress has been achieved, tracking objects under complex transformations remains a challenging problem for existing methods. While this year's top-performing methods have advanced in robustness through adaptive fusion and multi-stream reasoning, all trackers still fundamentally rely on memory-driven appearance matching. As such, their performance degrades under extreme transformations, fast motion, or near-identical distractors. Overcoming the current plateau may require a paradigm shift toward more explicit spatio-temporal modeling, or incorporation of external world priors beyond visual appearance alone.

The VOTSt2025 challenge winner. The top tracker according to the tracking quality score Q is DAM-RMemAOT (C.2) and thus the VOTSt2025 challenge winner.

4. Conclusion

The third VOTS2025 challenge results were presented. The challenge consolidate the tracking setups introduced in VOTS2023 [26] and introduce a real-time challenge to promote efficient architectures. Together, three categories were considered: the original VOTS2025 sub-challenge, which considers tracking of classical objects, its real-time extension VOTS-RT2025 and the VOTSt2025 sub-challenge introduced in 2024 that considers objects whose topology changes during tracking. All challenges use the same evaluation methodology and have reopened as benchmarks with public leaderboards for post-challenge submissions and evaluations.

Twenty trackers were evaluated in the VOTS2025 sub-challenge, including five baselines. The performance bar has been pushed substantially compared to the average performance in VOTS2023 and VOTS2024. As anticipated in VOTS2024 results presentation at ECCV2024, video-segmentation foundation model SAM2.1 [45] has made substantial impact on modern trackers. All top-performers are based on this foundation model and its recent extension DAM4SAM [51]. The winner of the VOTS2025 challenge nontrivially combines DAM4SAM and recent Cutie [7] and proposes modules for improving robustness against distractors, achieving substantial performance boost compared to other trackers. The boost primarily comes from the increased robustness, which enables the tracker to stay on the target longer.

Seven trackers were evaluated in the VOTS-RT2025 realtime sub-challenge and were re-run by the VOTS2025 committee on a common hardware for a fair comparison. The top performer on re-evalaution and VOTS-RT2025 winner was SV-DAM4SAM. This tracker extended DAM4SAM [51] by optimizing the feature extraction process and still using large (Hiera-L) backbone network, rather than using lightweight efficient networks. Interestingly, the realtime constraint did not significantly affect tracking robustness in top-performing trackers, but caused decrease in accuracy. This observation demonstrates a remarkable robustness of modern trackers, which are able to successfully track an object even if the input frames are frequently dropped.

The VOTSt2025 challenge observed a clear progress, primarily through adaptive fusion strategies. The top-

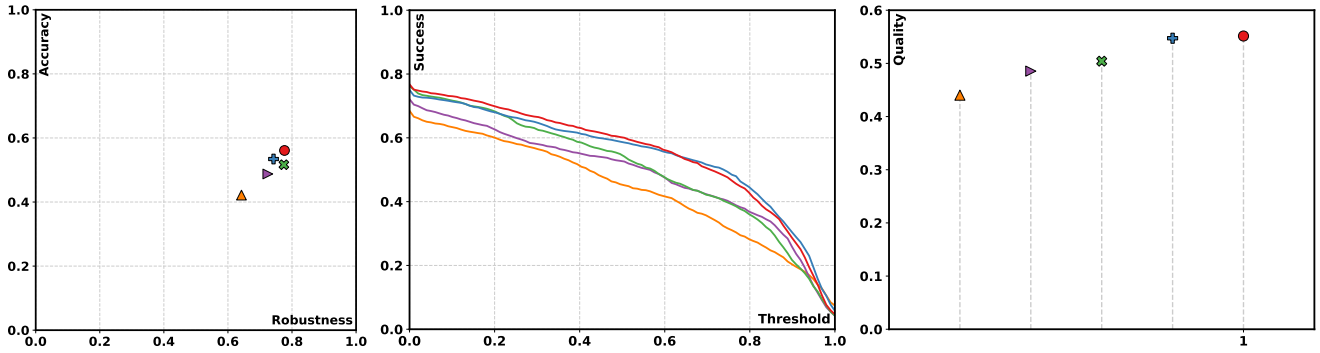


Figure 3. The VOTSt2025 challenge AR-plots (left), tracking quality Q-plots (center) and all trackers ranked according to Q score (right).

Tracker	Q	A	R	NRE	DRE	ADQ
DAM-RMemAOT	0.551①	0.561①	0.777①	0.086①	0.138	0.364
DQASOTt	0.547②	0.534②	0.742③	0.157	0.101①	0.557②
RMem	0.504③	0.517③	0.775②	0.092②	0.133③	0.341
AOTPlus	0.485	0.488	0.725	0.126③	0.150	0.404③
MMF_Tracker_T†	0.440	0.422	0.642	0.241	0.117②	0.598①

Table 4. Numerical results for VOTSt2025 challenge. Tracking quality (Q), accuracy (A), robustness (R), not-reported error (NRE), drift-rate error (DRE) and absence-detection quality (ADQ). $(\cdot)^{\dagger}$ denotes a tracker withdrawn by the authors after the deadline.

performing methods introduced mechanisms such as temporal diversity, confidence-aware prediction, and multi-modal fusion to better adapt to rapid appearance changes and complex motion. However, despite these improvements, their performance still plateaued under the most severe transformations — with the winning method reaching only 73% of the tracking quality score obtained by the winner on the VOTS2025 main challenge. This suggests that even sophisticated fusion of memory-driven appearance modeling with foundational tracking models like SAM2 is not sufficient to solve the task in its most difficult regimes. Robust tracking under extreme transformations remains an open challenge, likely requiring a shift toward more explicit spatio-temporal modeling and higher-level reasoning.

The VOT initiative continues to serve as the primary annual benchmarking activity for the general visual object tracking community. The challenges demonstrate the power of segmentation-based trackers, while also identifying room of improvements, which might particularly lie in lean, monolithic architectures that simultaneously achieve fast and robust tracking of solid and transformative objects. The future challenges will be steered based on the responses from the community and the results reported in this paper.

Statement of the co-authors contributions

The following abbreviations are used for the VOTS2024 organizers: Matej Kristan (MK), Jiří Matas (JM), Pavel Tokmakov (PT), Michael Felsberg (MF), Luka Čehovin Zajc (LČZ), Alan Lukežič (AL), Khanh-Tung Tran (KTT), Xuan-Son Vu (SV), Johanna Björklund (JB), Michal Neoral (MN), Hyung Jin Chang (HJC) and Gustavo Fernández (GF). The authors of sufficiently well performing trackers contributed in the public paper review and tracker descriptions editing. Their contributions to individual trackers are specified in Appendix A, Appendix B, and Appendix C.

Challenge coordination & oversight: MK, JM; Coordination of technical execution: AL; Technical support team: AL, MN, LCZ, KTT, MF, JB; Results interpretation: MK, PT, AL; Toolkit development: LCZ, AL; Paper drafting: MK, AL, PT, GF; Paper proofing: GF; Coordination of public review: GF; Camera ready preparation: GF; Evaluation server administration: KTT; Evaluation server team coordination: MF, JB; Website and design: LČZ; Sponsorship acquisition: MK, HJC.

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A. VOTS2025 submissions

This appendix summarizes the VOTS2025 challenge trackers and authors' contributions.

A.1. A Distractor-Aware Memory for Visual Object Tracking with SAM2 (DAM4SAM)

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DAM4SAM (Distractor-Aware Memory for SAM2) [51] is a memory-enhanced tracker built on top of the SAM2.1 foundational video segmentation model. It improves tracking robustness in the presence of distractors (visually similar objects) by introducing a dual-memory architecture: the Recent Appearance Memory (RAM) for maintaining accurate short-term tracking, and the Distractor Resolving Memory (DRM) for distinguishing the target from distractors. A novel update strategy leverages disagreements among SAM2's multiple output masks to detect distractors and selectively update DRM. This function-aware memory management effectively reduces drift and tracking failures.

A.2. A Unified Framework for High-Quality, Memory-Efficient, and Adaptively Robust Video Object Tracking (DQASOTb)

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We adopt a unified and self-contained tracking framework for all sequences in our video object segmentation pipeline. The method is designed to handle a wide range of video scenarios, including those with long durations, dynamic object appearances, and complex motion patterns, while remaining computationally efficient and highly adaptive. The tracker integrates three components that collectively enhance segmentation accuracy and tracking stability: (1) a high-quality mask refinement, (2) a selective memory mechanism and (3) a dynamic appearance modeling. Details of each component are explained in section A.12. For more details concerning the tracker, the reader is referred to [18].

A.3. A Unified Framework for Mask-Based Visual Object Tracking with SAM and DeAOT (DeSAMTrack)

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DeSAMTrack is a unified framework for mask-based visual object tracking that first utilizes the efficient tracking capabilities of DeAOT (Decoupling Features in Hierarchical Propagation) to generate coarse segmentation masks across video frames. These masks are then refined using the Segment Anything Model (SAM) [20], leveraging its powerful prompt-based segmentation ability to enhance mask accuracy and detail.

A.4. Adaptive Tracking via Selective Memory and Boundary Refinement for Sparse Object Scenarios (DQASOTg)

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For a subset of sequences, specifically those featuring long temporal spans, a small number of target objects and relatively large object sizes, we deploy a custom two-stage tracking framework builds upon the standard DMAOT pipeline. The core structure can be described in three components: (1) a high-quality mask refinement, (2) a selective memory mechanism and (3) a dynamic appearance modeling. Details of each component are explained in section A.12.

A.5. DAM4SAM With SAM1 (DAM4SAM2-SAM1)

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We first use DAM4SAM to generate masks for each frame, and then optimize them using SAM1.

A.6. DAMTrack base on DAM4SAM and Kalman filtering score metrics and DMAOT correction (DAMTrack)

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We propose a joint prediction tracker called DAMTrack, which is composed of DAM4SAM [51], DMAOT [26], and Kalman filter. The DAM4SAM proposes a new plug-in distractor-aware memory (DAM) and manageable strategy that substantially improves tracking robustness. We use DAM4SAM to track and segment each object while DMAOT can track multiple objects simultaneously and output their corresponding masks. We leverage the prediction results from the Kalman filter to compute Intersection over Union (IoU) with the outputs of both DAM4SAM and DMAOT. If the difference between these two scores is greater than a predefined threshold (equal to 0.4), it indicates that the two trackers have focused on different targets. In this case, the result with the higher score will be taken as the output. If the difference between the two scores is not significant, we assume that the two trackers are tracking the same object and the result of DAM4SAM, which enjoys higher accuracy, will be used as the output.

A.7. edgesam

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EdgeSam inference for video object tracking involves sequentially processing frames by first extracting hierarchical features with an efficient image encoder [5], then compressing dense frame-level memories from the memory bank using a 2D Spatial Perceiver that splits queries into global and patch-level groups to retain spatial structure while reducing computational load. The current frame's features serve as queries in memory attention blocks, fusing with compressed memories to integrate temporal context. Afterward, the mask decoder utilizes these features and prompt embeddings to predict the object mask. The new memory is encoded and enqueued into the bank in a FIFO manner.

A.8. Efficient Spatial Semantic trackerS (S3-DAM4SAM)

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We first introduce a proposed fusion block which utilizes the semantic and detailed information of the pretrained ViT models [42]. We fuse the information of the cls token from the ViT to multi-scale features and conduct local fusion between frame patches and multi-scale features for detailed fusion. In addition, to ensure the target representation of the target queries, we develop a discriminative query representation module in the query transformer to capture the local representation of the targets. The system features a shared image encoder that reduces computational overhead by 80% for multi-object scenarios, and employs precision-first decision-making based on IoU consistency, mask quality assessment, and temporal stability analysis. The system includes sequence-specific parameter optimization and comprehensive error handling, ensuring stable operation even when individual trackers fail.

A.9. Enhancing DMAOT with Test-Time Augmentation (DMAOT_TTA)

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We propose a robust test-time augmentation (TTA) ensemble pipeline. For each frame, we generate an ensemble of augmented views, primarily through horizontal flipping to address directional model biases, and multi-scale resizing to enhance performance on objects with varying scales. By introducing a rigorous state isolation mechanism we address the challenge of applying TTA to a stateful architecture like DMAOT. Before the TTA loop, a clean snapshot of the tracker's main memory engine is preserved. For each augmented view, a deep copy of this snapshot is created, forming a temporary, isolated inferences engine. All forward passes for a given view are performed on this disposable engine. The probability maps predicted from each isolated engine are transformed back to the original coordinate system and then averaged. This ensembling process produces a unified prediction.

A.10. Explicit visual prompts for visual object tracking (EVPTrack)

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This tracking system is built by combining EVPTrack [47] and the Segment Anything Model (SAM) [20]. EVPTrack serves as the primary tracker in our framework. Once EVPTrack identifies the object and provides a predicted bounding box, we employ the SAM model to perform segmentation. Its ability to produce high-quality segmentation masks based on simple prompts makes it a natural complement to the bounding boxes generated by EVPTrack. The overall pipeline works by first using EVPTrack to obtain accurate bounding box predictions for the target object, and then passing these bounding boxes to SAM, which generates precise segmentation masks accordingly.

A.11. Improving DMAOT Using HQ-SAM (HQ-DMAOT)

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The submitted HQ-DMAOT model is an extended version of the VOTS2023 winner tracker DMAOT [26], which introduces object-wise long-term memory to enhance the prediction accuracy of AOT [59] and DeAOT [60]. Based on DMAOT, the submitted HQ-DMAOT has one improvement: the introduction of pre-trained HQ-SAM [19] to refine tracking masks. Our approach is very straightforward: after obtaining multiple predicted masks from DMAOT in a single inference, we convert the masks into bounding boxes containing the targets. These bounding boxes are then fed into HQ-SAM along with the original image to obtain refined masks. We select the largest mask within each bounding box as the refined mask for the current target. In our implementation, we do not use the refined masks to update DMAOT. HQ-DMAOT jointly predicts and processes multiple tracking targets in a single inference. The final output of the HQ-DMAOT model is multiple masks refined by HQ-SAM.

A.12. Memory-Efficient, Boundary-Enhanced, and Language-Reinitialized Tracking for Long-Form Sparse-Object Video Segmentation (DQASOT)

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For video sequences characterized by extended duration, a small number of target objects, and relatively large object sizes, we replace the standard SAM2 pipeline with a specialized two-stage tracking framework built upon DMAOT. The core components of the framework are as follows: (1) Mask refinement: Initial object masks produced by the tracker are refined through an appearance-guided post-processing step. This stage improves boundary accuracy, removes spurious artifacts, and enhances segmentation details. (2) Selective memory management: To ensure scalability over long sequences, the tracker maintains a dynamic memory bank that retains only the most informative key frames. Each frame is scored based on its contribution to appearance diversity and temporal relevance. Redundant frames are pruned in real time. (3) Dynamic appearance updating: The appearance representation of each tracked object is periodically updated using the retained key frames. This allows the tracker to remain robust against occlusion, viewpoint change, and slow object deformation, while also minimizing drift over time. (4) Reinitialization via language-grounded detection: For very long sequences, we integrate Grounding DINO [48] with a text prompt to reinitialize the tracker. This language-guided mechanism allows the tracker to recover object alignment even after prolonged occlusion or ambiguous tracking states.

All other sequences in the dataset, typically shorter videos or those involving more complex multi-object interactions, are processed using the original SAM2 pipeline without modification.

A.13. Multi-modal Memory Fusion-based Tracker (MMF_Tracker)

NOTE: This tracker was withdrawn by the authors during the camera-ready phase and has been removed from the official VOTS online leader board.

MMF_Tracker_T combines SAM2 [45] and Depth-Anything-V2. SAM2 offers fine-grained, frame-wise object understanding, while Depth-Anything-V2 introduces depth-aware structural information. These models are fused through a learnable multi-scale fusion module to capture both appearance and spatial cues effectively. The tracker was trained on the Mega version of the CUTIE training [7].

A.14. SAMPT

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Our method employs an ensemble approach integrating SAM2 [45] and CoTracker [17]. Given the target object masks of the first frame, SAM2 predicts corresponding masks for subsequent frames. To address object deformation and occlusion in complex scenes, inspired by SAMURAI [56], we utilize CoTracker for inter-frame point tracking to optimize predicted mask selection. Given SAM2's predicted mask of previous frame as input, CoTracker infers current positions of foreground points and their visibility flags. Then, the IoU score between foreground points and the predicted masks of current frame from SAM2 is calculated. The final selected mask maximizes a weighted combination of this IoU score and SAM2's original affinity score. The visible keypoints then serve as point prompts for SAM2, which are fed into the prompt encoder to refine the mask. This mechanism guides mask refinement toward structurally significant regions with consistent trajectories [17]. By integrating CoTracker into SAM2, our framework addresses dynamic scene challenges through occlusion-aware mask selection and keypoint-guided refinement.

A.15. MIEM-HSE_ALPHACHIP_Qwen_VL

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The key innovation of our tracker lies in its zero-shot capability to process diverse objects without task-specific training while maintaining optimal computational efficiency and tracking accuracy. We did not train additional models but only utilized pre-existing solutions not originally designed for tracking tasks. To find an object matching the template in a frame position close to the given one, we use the multimodal model Qwen2.5-VL (7B variant) [2]. This model, based on a specific query, returns one or several Bounding Boxes for a given frame. If no object is found, it returns a message indicating the absence of the object. The mask inside the Bounding Box is extracted using the Segment Anything Model 2 (SAM2) [20]. If multiple candidate objects are detected, a simple heuristic approach is applied to calculate the candidate's weight based on: the position of the object from the previous frame; the position of the candidate bounding box; the distance between the centers of the boxes; the ratio of their areas; and the IoU (Intersection over Union) for the objects.

B. VOTS-RT2025 submissions

This appendix summarizes the VOTS-RT2025 challenge trackers and authors' contributions.

B.1. A Lightweight Distractor-Aware Memory for Visual Object Tracking with SAM2 (DAM4SAMLight)

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This tracker is built upon the vision foundation model SAM2. Unlike conventional nearest neighbor-based memory management strategies, DAM4SAM not only leverages both the nearest neighbor and the most reliable first-frame to provide robust target appearance information, but also incorporates hard example mining. Specifically, it takes into account frames containing background distractors with similar appearance to the foreground, which allows the model to be informed of the existence and spatiotemporal positions of potential distractors, thereby preventing tracking drift. In total, we utilize 7 memory frames to establish temporal attention associations. To achieve real-time tracking performance, we experimented with lower input resolutions and more lightweight backbone architectures, such as HieraViT-Tiny.

B.2. Adaptive Tracking via Selective Memory and Boundary Refinement for Sparse Object Scenarios (DQASOTrg)

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For a tracker description, the reader is referred to section A.4.

B.3. Adaptive Video Object Segmentation via Tracker Selection Between Memory-Augmented and Segment-Anything Models (DQASOTr)

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This tracker adopts a dual-tracker strategy based on the characteristics of the input sequence. Specifically, for sequences exhibiting longer temporal spans, fewer objects (typically one or two), and relatively large target sizes, we employ DMAOT in place of the standard SAM2 pipeline. In contrast, for the remainder of the dataset—typically composed of shorter clips, more frequent scene changes, or multiple interacting objects—we retain the SAM2 pipeline due to its computational efficiency and strong per-frame segmentation accuracy. By aligning the choice of tracker with the temporal and structural characteristics of each video, we observe notable gains in both boundary quality and mask stability for long, sparse-object sequences. The overall system remains efficient and adaptable, balancing SAM2's speed with DMAOT's robustness in extended tracking scenarios. This selective integration enables us to leverage the strengths of both models without retraining or introducing additional architecture overhead.

B.4. Associating Objects with Multi-scale Transformers for Video Object Segmentation (MSAOT)

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The MSAOT tracker is built based on AOT [58, 59, 61], a transformer-based video object segmentation method, by applying transformers in multiple feature scales. MS-AOT tracks and segments most of the objects end-to-end without using bounding-box information. For tiny objects, we use MixFormer [8], a bounding-box-based tracker, to coarsely locate the objects before applying MS-AOT to predict segmentation results. The backbone of MS-AOT is ResNet-50, and the backbone of MixFormer is CvT [53].

B.5. edgesam (EdgeSAM)

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For the tracker description the reader is referred to A.7.

B.6. Efficient Spatial Semantic trackerS (SV-DAM4SAM)

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Our approach builds on the state-of-the-art long-term tracker DAM4SAM, which enhances SAM2's tracking capabilities via a distractor-aware memory (DAM) framework. DAM splits memory into two components: Recent Appearance Memory (RAM), which captures the most recent target appearances for accurate segmentation, and Distractor-Resolving Memory (DRM), which stores carefully selected anchor frames to distinguish the true target from visually similar distractors. A novel introspection-driven update mechanism monitors disagreements between the predicted and alternative masks—when divergence is detected, new frames are added to the DRM, thereby boosting resilience in cluttered or occluded scenes without requiring additional training. A key innovation is our parallel feature memory design: in a single encoder pass per frame, we extract features and simultaneously write them into both RAM and DRM, enabling multi-object inference in parallel with just one forward computation. GPU memory usage is managed through strategic cache clearing and efficient tensor operations. This multi-layered approach ensures the tracker can handle extended video sequences without memory-related performance degradation, making it suitable for real-time applications and indefinitely long tracking tasks.

B.7. LowFormerTrack MOT TAM (LowFormer)

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LowFormer-Track is an adaptation of the Separable Self and Mixed Attention-based Tracker (SMAT) [13]. We improved its Speed-Accuracy trade-off by replacing its backbone with the LowFormer-B15 vision backbone [41] and applied LowFormer-Attention in the predictor head instead of Separable Self-Attention. LowFormer-Track achieves real-time performance on embedded hardware, with 51 fps on the Nvidia Jetson TX2, while having an AUC of 78.9 in GOT10K [16] and 61.7 on LaSOT [11]. Finally, we use Efficient Track Anything Model (EfficientTAM) [55] to convert the bounding box to a segmentation mask.

C. VOTSt2025 submissions

This appendix summarizes the VOTSt2025 challenge trackers and authors' contributions.

C.1. AOT+ (AOTPlus)

Submitted by VOTS committee

AOTPlus [50] extends the AOT approach [59] by increasing the length of the training sequences, replacing the short-term memory module with a recurrent transformer and increasing the temporal resolution of the model at test time.

C.2. Efficient Spatial Semantic RemAOTS (DAM-RMemAOT)

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DAM-RMemAOT is a video object segmentation framework that builds upon the original RMemAOT architecture [63]. The model employs a parallel multi-tracker framework consisting of a main tracker with data augmentation support, two temporally offset auxiliary trackers [63], and an on-demand emergency recovery tracker. Each tracker is equipped with an intelligent confidence estimation module that evaluates tracking quality through prediction consistency analysis and mask quality assessment. We replace fixed IoU thresholds with dynamic, confidence-aware decision making. The system calculates combined scores (IoU × Confidence × Scene Context) to select optimal predictions, enabling context-aware selection and graceful degradation when primary trackers struggle. An adaptive Threshold Management dynamically adjusts fusion parameters based on scene complexity estimation, historical performance feedback, and object-specific characteristics. When all trackers fail simultaneously, the system activates recovery protocols using historical mask analysis, morphological expansion operations, and progressive recovery strategies to maintain tracking continuity. A temporal diversity strategy initializes auxiliary trackers at different temporal offsets (frames 1 and 2), creating diverse memory banks that capture different aspects of object evolution.

C.3. Multi-modal Memory Fusion-based Tracker (MMF_Tracker_T)

NOTE: This tracker was withdrawn by the authors during the camera-ready phase and has been removed from the of-

ficial VOTS online leader board.

MMF_Tracker_T combines SAM2 [45] and Depth-Anything-V2. SAM2 offers fine-grained, frame-wise object understanding, while Depth-Anything-V2 introduces depth-aware structural information. These models are fused through a learnable multi-scale fusion module to capture both appearance and spatial cues effectively. The tracker was trained on the Mega version of the CUTIE training [7].

C.4. RMem: Restricted Memory Banks Improve Video Object Segmentation (RMem)

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The method proposed in RMem focuses on enhancing video object segmentation (VOS) by restricting the size of memory banks. This approach diverges from the conventional practice of continuously expanding memory banks to include more historical frames. Our key insight is that larger memory banks actually complicate the decoding process due to redundant and less relevant information. More focused memory banks enable VOS modules to better decode relevant features. Furthermore, this approach reduces the discrepancy between training and inference stages, as VOS models are usually trained on short clips but applied to longer videos during inference. Our method, called RMem, includes strategies to update the memory bank efficiently, balancing relevance and freshness of frames. We draw inspiration from the upper confidence bound algorithm [1], traditionally used in multi-arm bandit problems, to manage this balance. Additionally, the restricted memory approach facilitates the use of temporal positional embeddings, improving spatio-temporal reasoning by encoding the order of frames. We highlight the significant improvements in both overall VOS quality and robustness to object state changes due to this memory restriction strategy.

C.5. Spatio-Temporal Refinement and Integration for Dynamic Environments (DQASOTt)

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The tracker integrates three components that collectively enhance segmentation accuracy and tracking stability: (1) a high-quality mask refinement, (2) a selective memory mechanism and (3) a dynamic appearance modeling. Details of each component are explained in section A.12. To further strengthen temporal modeling capabilities, we train our DeAOT-based model on the VOST dataset [50] using a Swin Transformer encoder [18], which provides a powerful representation of spatial features. We also incorporate Temporal Positional Embeddings from RMEM [63], allowing the model to more effectively capture motion patterns and temporal ordering across frames.

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